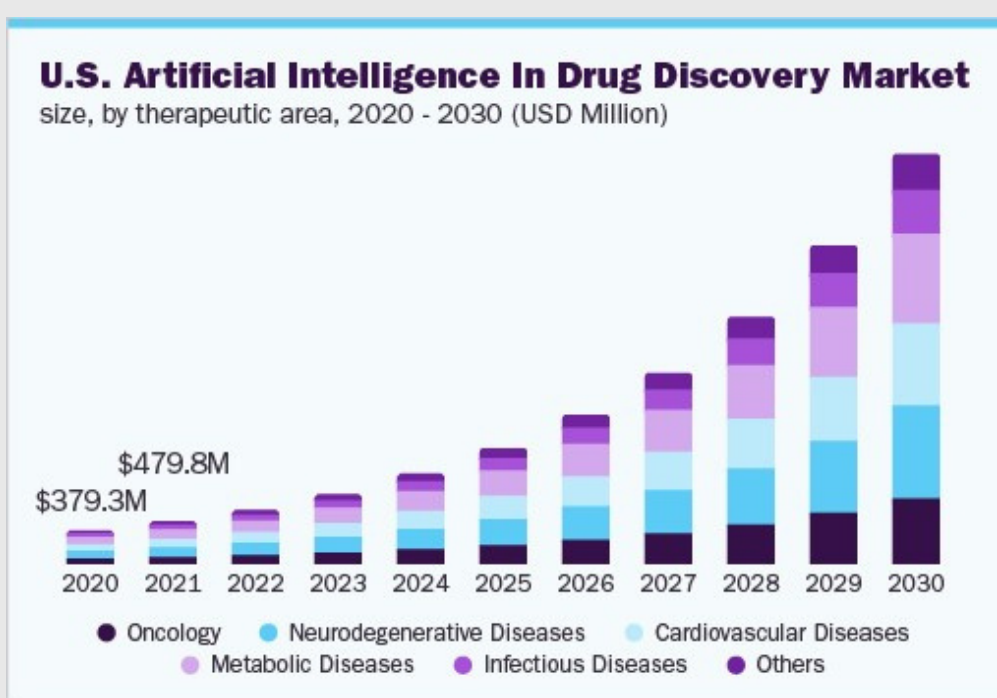


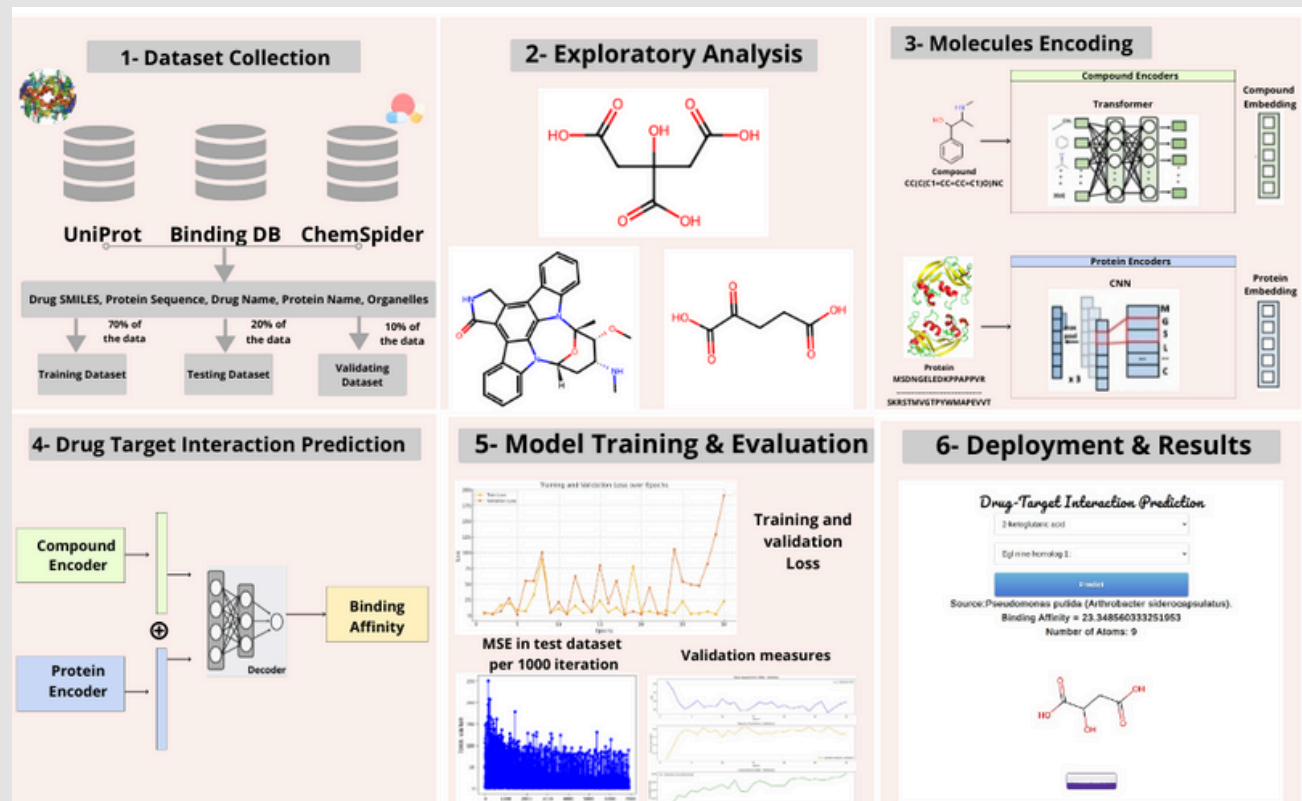
Motivation

- The global AI in drug discovery market is set to expand at a remarkable CAGR of 29.6% from 2023 to 2030, heralding a transformative era in therapy development.
- Our project focuses on drug-target interaction based on binding affinity, using AI to revolutionize this critical process. AI streamlines drug discovery, reduces costs, and accelerates timelines by enhancing the accuracy of molecular interaction predictions.
- This technology drives unprecedented innovation in life sciences, enabling rapid development of targeted treatments and improving health outcomes. By analyzing vast datasets and predicting binding affinities, AI opens new frontiers in understanding diseases and developing precise therapies.
- The future of medicine is being redefined by AI, where innovation meets efficiency, paving the way for a healthier, advanced world.



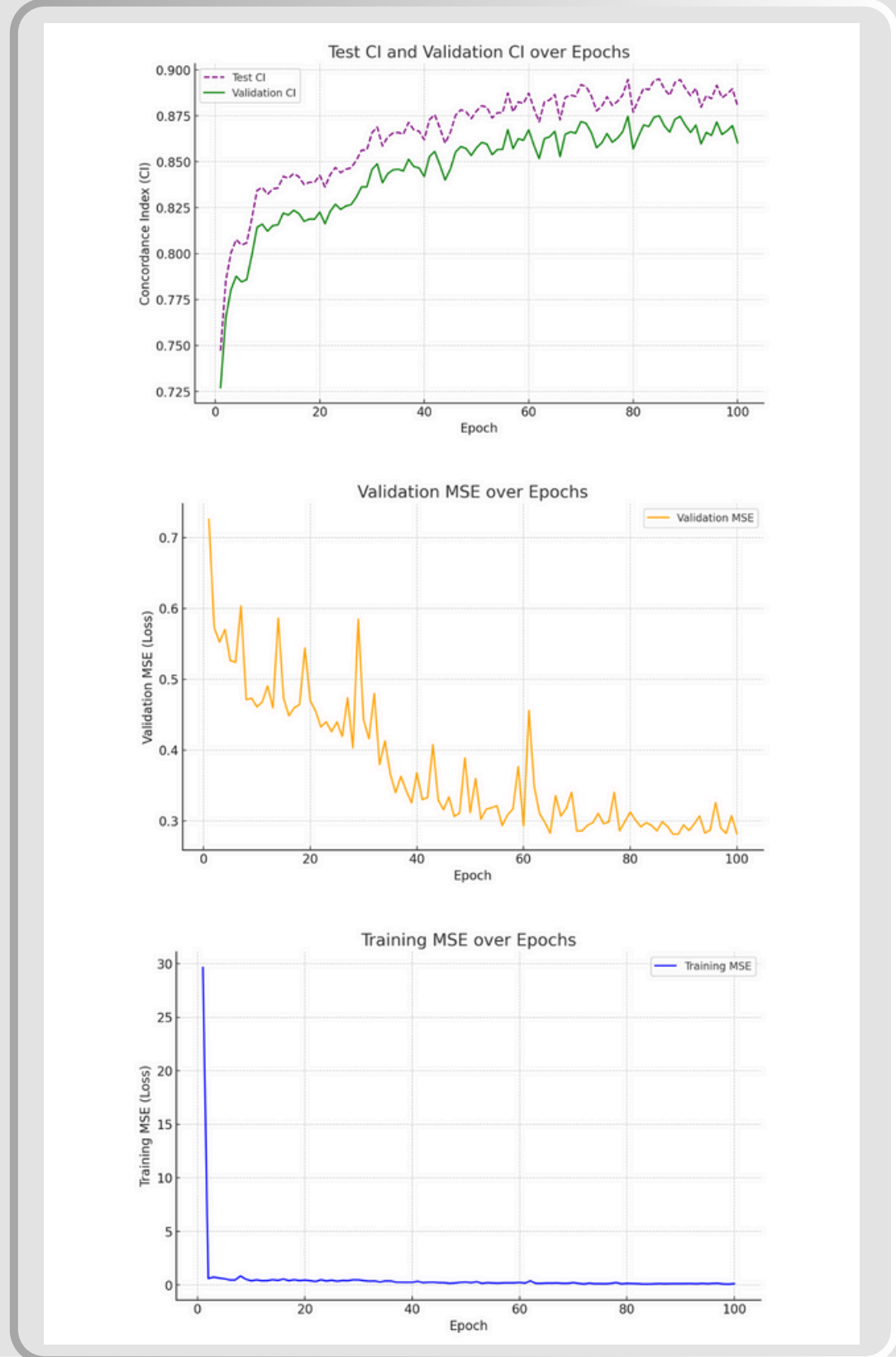
Objective

our project aims to develop Deep-Learning model that can accurately predict the binding affinities between small-molecule drugs and their target proteins. By achieving this, the project aims to significantly reduce pharmaceutical research costs by minimizing the reliance on high-throughput screening. Additionally, it seeks to enlarge the search space for potential drug candidates, thereby avoiding the risk of missing promising compounds.



Results and analysis

- we present the results and analysis of our drug-target interaction (DTI) prediction model. To evaluate the performance of our model, we used several key metrics: training and validation loss over epochs, mean squared error (MSE), concordance index, and Pearson correlation. providing a comprehensive assessment of the model's predictive accuracy and reliability

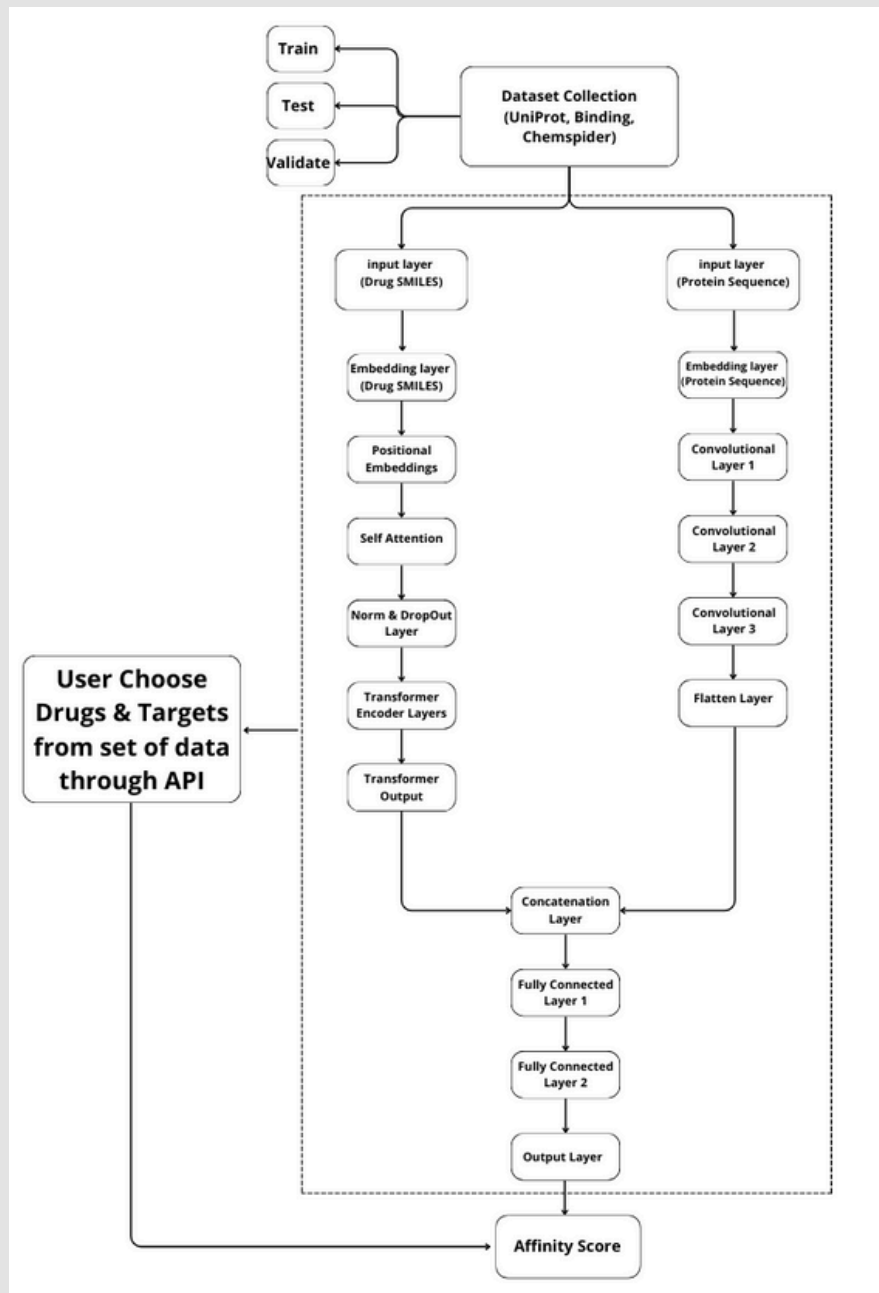


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Materials and methods

- Data Collection and Cleaning**
 - Data from Binding Database via TDC Library includes drug and target IDs, SMILES, sequences, and binding affinity scores. Names were retrieved from ChemSpider and UniProt. Cleaning involved removing null values, outliers, and scaling the target variable.
- Data Exploratory Analysis**
 - EDA used RDKit to generate molecular structures and analyze atom counts in drug molecules, aiding dataset understanding.
- Model Construction and Evaluation**
 - The model predicts binding affinity by processing drug and target sequences through embedding layers, positional embeddings, convolutional layers, and transformer encoder layers.
- Model Deployment**
 - Using Flask, HTML, and CSS, the model offers a web interface and API for predictions, enabling biologists to predict drug-target interactions, thus speeding up research and decision-making.



Implementation

- After applying our methods:
 - Data collection and Cleaning
 - Data Exploratory Analysis
 - Model Construction and Evaluation
 - Model Deployment
- we had to deploy the model to enable the biologists to measure the binding affinity score between drug and protein.

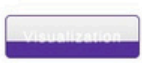
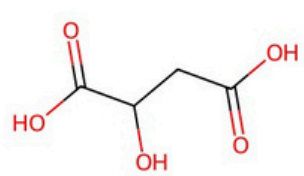
Drug-Target Interaction Prediction

2-ketoglutaric acid

Egl nine homolog 1;

Predict

Source:Pseudomonas putida (Arthrobacter siderocapsulatus).
Binding Affinity = 23.348560333251953
Number of Atoms: 9



Future work

Future work for this DTI project will focus on enhancing the accuracy of our AI models by:

- integrating advanced deep learning techniques, such as transformers and graph neural networks.
- incorporating structural and omics data to provide a more comprehensive understanding of drug-target interactions.
- Expanding the model's applicability to new drug formats and cross-species analysis.
- integrating clinical trial data and exploring personalized medicine applications to ensure the model's relevance to real-world scenarios.

Conclusion

- Our model predicts binding affinity for various drugs and target proteins using only structural information of compounds and amino acid sequences of proteins. This "in silico" approach can streamline drug discovery, offering insights into drug efficacy and interactions without extensive experimental data.

